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Fenestration unit having integrated power interface for peripheral devices

Abstract

A fenestration unit for a dwelling is disclosed. The fenestration unit, in the form of a window unit or a door unit includes a frame supporting a movable panel. A power supply module includes a battery pack disposed in one of the frame and the moveable panel. A power outlet module is also mounted in one of the frame and the movable panel. The power outlet module has a power outlet connector electrically coupled to the battery pack and is configured to resist water penetration therein. A homeowner device may be electrically coupled to the power outlet connector such that the power supply module powers the homeowner device.

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) The present disclosure relates to fenestration units and more particularly to fenestration units configured with power inlet modules, power outlet modules and ancillary components including peripheral devices for the fenestration unit.

BACKGROUND OF THE INVENTION

(2) This section provides background information related to the present disclosure which is not necessarily prior art.

(3) Significant advances have been made in the area of home computing networks over the past decade. As a result there has been a proliferation of peripheral devices for automating functions associated with buildings and in particular dwellings. Such devices include lights, power outlets, thermostats, and window shades, as well as a variety of appliances and electronics. Such device may also include various sensors—light sensors, motion sensor, temperature sensors, and the like—for measuring environmental condition that may be used in more sophisticated home automation systems. It should be appreciated that many of these types of devices may be located on or near a fenestration unit.

(4) While many of these devices connect wirelessly to the home computing network, they often require an external power source for operating the device and/or charging batteries of the device. The external power source may be hardwired to the device or alternately connected to a conventional electrical outlet via a transformer block with a standardized cable such as a USB cable. In either case, powering these devices can require a significant investment in time and/or money to hardwiring them or can create unsightly wiring running from the nearest electrical outlet to the device.

(5) Accordingly, there is a need in the art of fenestration units to provide an integrated interface for powering such peripheral devices.

SUMMARY OF THE INVENTION

(6) This section provides a general summary of the disclosure and is not a comprehensive disclosure of its full scope or all of its features.

(7) The present disclosure is for a fenestration unit with a built-in battery and a power outlet

module operable to power homeowner devices that may include security cameras, flood lights, interior or exterior sensors of a dwelling, or other similar devices of homeowner's needs. The fenestration unit includes an in-situ power supply module and a power outlet module. The power supply module may be permanently installed in the fenestration unit or alternately may be removably disposed in the fenestration unit. The power outlet module may include a standardized output jack that is accessible on the interior or the exterior of the fenestration unit. When accessible on the exterior of the fenestration unit, the power outlet module has weather-resistant features to ensure proper operation and durability of the module. Such weather-resistant features may include a silicone enclosure (i.e., cap and/or wire connectors) for the output jack and/or a downwardly-angled orientation of the output jack for preventing water intrusion (e.g., IPX4) and permitting drainage if water or water vapor built-up occurs.

(8) The power supply module specifications may be configured according to the power requirements and specifications for the desired applications and the device(s) to be operated. The power supply module may be a re-chargeable power supply module that can be re-charge using external or internal power sources. An external power source may include home electrical power outlet. In the case of external power, the power supply module may include a power inlet module having a standardized input jack that is accessible on the interior of the fenestration unit and a power transformer configure to convert the source power (e.g., 110-120 VAC in the US) to a power compatible for the battery. An internal power source may include solar power generated using transparent photovoltaic glass (e.g. perovskite solar cell; organic solar cells) or other types of solar cell integrated into the non-glass structure of a fenestration unit. In the case of solar power, the power supply module may include a power conditioning module configure to condition the solar-generated power to a power compatible for the battery.

(9) Further areas of applicability will become apparent from the description provided herein. The description and specific examples in this summary are intended for purposes of illustration only and are not intended to limit the scope of the present disclosure.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The drawings described herein are for illustrative purposes only of selected embodiments and not all possible implementations and are not intended to limit the scope of the present disclosure.

(2) FIG. 1 shows an interior side of a window in accordance with the present disclosure;

(3) FIG. 2 shows an exterior side of the window shown in FIG. 1;

(4) FIG. 3 shows an interior side of a door in accordance with the present disclosure;

(5) FIG. 4 shows an exterior side of the door shown in FIG. 3;

(6) FIG. 5 shows a detail of an integral power supply module;

(7) FIG. 6 shows a detail of an accessible power supply module;

(8) FIG. 7 shows a schematic circuit diagram of the integrated power interface and a homeowner device;

(9) FIG. 8 shows a schematic cross-section showing the fenestration unit in accordance with the present disclosure; and

(10) FIG. 9 shows an exterior side of a window having an insulated glass unit with photo-voltaic solar devices.

(11) Corresponding reference numerals indicate corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

(12) Example embodiments will now be described more fully with reference to the accompanying drawings.

(13) With reference to the figures, a fenestration unit **10** with a built-in power supply module **40** and a power outlet module **60** operable to power a homeowner device **80**. With particular reference to FIGS. **1** and **2**, the fenestration unit **10** is in the form of a double hung window unit **12**. One skilled in the art should understand that the double hung window unit **12** represents an exemplary embodiment and that the present disclosure contemplates other types of window units including but not limited to single hung, casement, awning, hooper, sliding, or fixed glass units, adapted with a built-in power supply module and a power outlet module in a manner similar to that described herein. The double hung window unit **12** includes a window frame **14** slidably supporting a lower sash **16** and an upper sash **18**.

(14) FIG. **1** shows the interior side of the window unit **12** having one or more power supply modules **40.1**, **40.2**, **40.3** built into a component for the window unit **12**. For illustrative purposes, power supply module **40.1** is shown built into the window frame **14**, power supply module **40.2** is shown built into a lower sash rail **22** and power supply module **40.3** is shown built into an upper sash rail **24**. While multiple power supply modules are illustrated, one such unit may be sufficient for a given application. One skilled in the art should understand that a power supply module may be built into other locations on the window unit **10**. These power supply module **40.1**, **40.2**, **40.3** may be singularly or collectively referenced herein as a power supply module **40**.

(15) FIG. **2** shows the exterior side of the window unit **12** having one or more power outlet modules **60** built into a component of the window unit **10**. For illustrative purposes, power outlet module **60.1** is shown built into the window frame **14**, power outlet module **60.2** is shown built into the lower sash rail **22**, and power outlet module **60.3** is shown built into the upper sash rail **24**. While multiple power outlet modules are illustrated, one such unit may be sufficient for a given application. One skilled in the art should understand that a power outlet module may be built into other locations on the window unit **12**. These power outlet modules **60.1**, **60.2**, **60.3** may be singularly or collectively referenced herein as a power outlet module **60**.

(16) With reference now to FIGS. **3** and **4**, the fenestration unit **10** is in the form of a single-hinged entry door unit **26**. One skilled in the art should understand that the single-hinged entry door unit represents an exemplary embodiment and that the present disclosure contemplates other types of door units, including but not limited to double-hinged entry or sliding door units, adapted with a built-in power supply module and a power outlet module in a manner similar to that described herein. The single-hinged door unit **26** includes a door frame **28** hingedly supporting a door slab **30**.

(17) FIG. **3** shows the interior side of the door unit **26** having one or more power supply module **40.4**, **40.5**, built into a component for the door unit **26**. For illustrative purposes, power supply module **40.4** is shown built into the door frame **28**, power supply module **40.5** is shown built into the door slab **30**. While multiple power supply modules are illustrated, one such unit may be sufficient for a given application. One skilled in the art should understand that a power supply module may be built into other locations on the door unit **26**. These power supply module **40.4**, **40.5** (along with **40.1**, **40.2**, **40.3**) may be singularly or collectively referenced herein as a power supply module **40**.

(18) FIG. **4** shows the exterior side of the door unit **26** having one or more power outlet modules **60** built into a component of the door unit **26**. For illustrative purposes, power outlet module **60.4** is shown built into the door frame **28** and power outlet module **60.5** is shown built into the door slab **30**. While multiple power outlet modules are illustrated, one such unit may be sufficient for a given application. One skilled in the art should understand that a power outlet module may be built into other locations on the door unit **26**. These power outlet modules **60.4**, **60.5** (along with **60.1**, **60.2**, **60.3**) may be singularly or collectively referenced herein as a power outlet module **60**.

(19) With reference now FIG. **5**, the power supply module **40** may include a battery pack **42** that is concealed within an interior volume of the fenestration unit **10** such that the battery pack **42** is not accessible once the fenestration unit **10** is installed in an opening (not shown). Alternately, as

shown in FIG. 6, the power supply module **40'** may include a battery pack **42** that is concealed within an interior volume of the fenestration unit **10**, which is enclosed by a cover plate **44** that is releasably secured to the fenestration unit **10** to provide access to the battery pack **42** once installed in an opening (not shown). In one embodiment, the battery pack **42** includes one or more batteries **42.1, 42.2** that are rechargeable by a power source **46**. The batteries **42.1, 42.2** may be any suitable type of rechargeable battery such as Alkaline, Ni—Cd, Li-ion, Ni-MH, Li-polymer or Zn—C for example. As best seen in FIGS. 5 and 6, the power supply module **40** may include a female power connector **48** for electrically coupling a power source **46** to the power supply module **40**.

(20) The power source **46** may be supplied from the electrical system in the building that the fenestration unit **10** is installed. The fenestration unit **10** may be hard-wired or externally coupled via a wall outlet to the electrical system. The power source **46** may be a solar panel array supported on the fenestration unit **10** as further detailed below. The power source **46** may also include a combined power supply from the electrical system and the solar panel array **100, 110, 120**.

(21) With reference now to FIG. 7, the power supply module **40** may include an input-side power transformer **50** configured to convert and condition the input power from the power source **46** to match the charging power requirements for the battery pack **42**. The power transformer **50** may be external of the fenestration unit **10** such as a power adapter or “wall wart” plugged into a wall outlet (not shown) for converting the input voltage from the power grid (e.g., 110 VAC in USA) to a suitable charging voltage (e.g., 1-10 VDC) for the battery pack **42**. Alternately, the power transformer **50** may be internal of the fenestration unit **10** such as a power conditioning circuit **50** for converting the input voltage, whether from the power grid (e.g., 110 VAC in USA) or from a solar panel array (e.g., 12-24 VDC) to a suitable charging voltage (e.g., 1-10 VDC) for the battery pack **42**.

(22) The power supply module **40** may include a microprocessor **52** electrically coupled to the battery pack **42**. The microprocessor **52** may be programmed for controlling various functions of the power supply module **40** including regulating the input voltage and output voltage within a proper operating range, detecting operating parameters of the power supply module **40** and providing status signaling to an indicator display **54** (see FIGS. 5 and 6) on the power supply module **40**. The microprocessor **52** may also monitor an on-board sensor **56** in the fenestration unit **10**. The sensor **56** may be configured to sense various environmental conditions around the fenestration unit **10** such as light, temperature or humidity. The microprocessor **52** may also be configured with wireless communication capabilities over various wireless networks (e.g., WLANS, WPANS, and/or WMANS as well as M2M, P2P or other distributed ledger technology). With such communication capabilities, the fenestration unit **10** may be configured as a component within a home automation network. The microprocessor **52** is configured to operate in a variety of network environments with various software interfaces including macOS, Windows, Linux, iOS, Android and similar standard operating systems.

(23) The power supply module **40** may include an output-side power transformer **58** configured to convert and condition the power from the battery pack **42** to match the power requirements of the power outlet module **60**. (e.g., at least 200 mAh, 5-20 VDC). As illustrated in FIGS. 7 and 8, the power supply module **40** is electrically coupled to a female output connector **62** in the power output module **60**. Any suitable type of connector may be employed as the power output connector **62**, for example USB-A, USB-B, USB-C, micro USB, Thunderbolt, 9v, HDMI, CAT-POE or other conventional connectors used with homeowner devices. While a single output connector **62** is illustrated in FIG. 8, one skilled in the art should appreciate that the power outlet module **60** may include one or more outlet connectors of the same or different configurations.

(24) As mentioned above, the input power for the power supply module **40** may be a solar panel array configured in various means and supported on the fenestration unit **10**. With reference to FIGS. 1-4, one embodiment of a solar panel array **100** includes a plurality of solar cell **102** supported on the frame **14, 28** of the fenestration unit **10**. In another embodiment, a solar panel

array **110** includes a plurality of translucent solar cells **112** (shown in broken lines) applied on the surface (e.g., glass panes or door panels) of the fenestration unit **10**. FIG. **9** illustrates another embodiment in which a solar panel array **120** include a plurality of solar cell **122** supported on a spacer edge **124** of the insulated glass unit or IGU **126**. In particular, a first solar array **120.1** includes a plurality of solar cells **122.1** (shaded region) disposed in the IGU **126** supported in the lower sash **16** and a second solar array **120.2** includes a plurality of solar cells **122.2** (shaded region) disposed in the IGU **126** supported in the upper sash **18**. One skilled in the art should appreciate that a fenestration unit **10** may include one or more of these solar panel arrays, **100**, **110**, **120**. Wiring (not shown) is disposed within the fenestration unit **10** and electrically couples the solar panel array(s) **100**, **110**, **120** to the power supply module **40**.

(25) reference now to FIG. **8**, a power outlet module **60** configured for exterior exposure is further described. The power outlet module **60** includes a weatherproof housing **64** disposed within a structural component **66** of the fenestration unit **10**. The structural component may be a frame **14**, **28**, a sash rail **22**, **24** or a door slab **30**. The female connector **62** is oriented in the housing **64** in an angularly downward direction a for providing natural draining of any water that may collect in the connector **62**. For example, the angular orientation a may be in the range of 5-20°. The weatherproof housing **64** is configured to resist water intrusion from environmental condition caused by weather (e.g., drizzle, rain, snow, sleet, ice, etc.) or water spraying equipment (e.g., sprinklers, hoses, sprayers, etc.) The weatherproof housing **64** includes a flexible silicone grommet **68** disposed over the opening of the connector **62** to inhibit water or water vapor from entering the connector **62**. An access slot **70** formed in the grommet **68** is positioned over the female connector **62** for allowing a complementary male connector **72** to be inserted into the female connector **62**. The male connector **72** may include a seal **74** which engages the female connector **62** to inhibit water penetration into the power outlet module **60**. Other means for weatherproofing include covers, seals, sockets, plugs, housings, and the like which may be implemented to provide a weatherproof housing **64** for the power outlet module **60**. In one embodiment, the power outlet module **60** provides protection of the connector **62** from spraying or splashing of water (or other liquids) in all direction such that its level of resistance to water ingress is rated at IPX4 or higher under the Ingress Protection standard. A homeowner device **80** is coupled to the power supply module **40** via the power outlet module **60** when the male connector **72** is inserted into the female connector.

(26) From the foregoing detail description it is apparent that the present disclosure provides a fenestration unit (e.g., window and/or door units) with built-in battery and input/output ports for powering homeowner devices that include but are not limited to security cameras, flood lights, sensors inside or outside of home, and other devices of homeowner's needs. The fenestration unit may include a battery pack that can be charged with home power outlet (e.g., ~110V in USA and ~220V in China), built-in transparent photovoltaic glasses (e.g., perovskite solar cell; organic solar cells), or other types of solar cell integrated into the fenestration unit (e.g., solar cell that are put onto the spacer).

(27) The battery capacity may be customized to desired applications and are hid inside the frame of the fenestration unit. Battery management components are provided to control the charging and discharging of the battery at optimal conditions regardless of temperature. The weatherproofing of battery component inside the fenestration unit may be provided with the use of conformal coating, which can be apply using different application methods including spray coating, dip coating, brush coating and etc. The composition of the conformal coating can be a single use or combined use of acrylics, silicones, urethanes, perylenes and other suitable polymerics that prevent moisture, dust, temperature extremes and chemicals from damaging the performance and lifetime of the electrical components The electrical components are waterproofed to a rating of IPX4 for providing durability upon contact with water or water vapor.

(28) The input/output ports can be one or more in numbers and can be fabricated at different

location on the fenestration unit. The input/output port may be tilted to provide channels for water drainage if there is a water built-up inside the port. Moreover, the ports can be covered with silicone cap when not in use and can be covered with silicone connectors when plugged-in with homeowner devices.

(29) The foregoing description of the embodiments has been provided for purposes of illustration and description. It is not intended to be exhaustive or to limit the disclosure. Individual elements or features of a particular embodiment are generally not limited to that particular embodiment, but, where applicable, are interchangeable and can be used in a selected embodiment, even if not specifically shown or described. The same may also be varied in many ways. Such variations are not to be regarded as a departure from the disclosure, and all such modifications are intended to be included within the scope of the disclosure.

Claims

1. A fenestration unit for a dwelling comprising: a frame supporting a movable panel; a power supply module including a battery pack disposed in one of the frame and the moveable panel; and a power outlet module mounted in an exterior surface of at least one of the frame and or the movable panel, the power outlet module having a weatherproof housing supporting a power outlet connector electrically coupled to the battery pack, wherein the weatherproof housing includes a flexible grommet disposed on the exterior surface over the power outlet connector and providing a weather resistance rating of at least IPX4, the flexible grommet having an access slot formed therethrough to allow a complementary power connector to be inserted through the flexible grommet and into the power outlet connector.
2. The fenestration unit of claim 1, wherein the weatherproof housing supports the power outlet connector in an angularly downward orientation.
3. The fenestration unit of claim 1, wherein the power supply module comprises a rechargeable battery pack.
4. The fenestration unit of claim 3, further comprising a power source operable coupled to the rechargeable battery pack.
5. The fenestration unit of claim 3, further comprising a power transformer operable coupled between the power source and the rechargeable battery pack for converting power from the power supply to power suitable for charging the battery pack.
6. The fenestration unit of claim 5, wherein the power source comprises an electrical system powering the dwelling and the power transformer is operable to convert power from the electrical system to power for recharging the battery pack.
7. The fenestration unit of claim 5, wherein the power source comprises a solar panel array supported on the fenestration unit and the power transformer is operable to convert power from the solar panel array to power for recharging the battery pack.
8. The fenestration unit of claim 7, wherein the solar panel array comprises a plurality of solar cells supported on the frame of the fenestration unit.
9. The fenestration unit of claim 7, wherein the solar panel array comprises a plurality of solar cells supported on the movable panel of the fenestration unit.
10. The fenestration unit of claim 7, wherein the power source further comprises an electrical system powering the dwelling and the power transformer is operable to convert power from the electrical system to power for recharging the battery pack.
11. The fenestration unit of claim 3, further comprising a power conditioning circuit operably coupled between the battery pack and the power outlet module for converting power from the battery pack to power suitable for a homeowner device configured to be connected to the power outlet connector.
12. The fenestration unit of claim 3 further comprising a microprocessor operable coupled to the

battery pack and configured to regulate at least one of an input voltage to the battery pack and an output voltage from the battery pack.

13. The fenestration unit of claim 12, further comprising a status indicator, wherein the microprocessor is configured to monitor a state of at least one of the power supply module and the power outlet module and to communicate the state to the status indicator.

14. The fenestration unit of claim 12, further comprising a sensor for measuring an environmental parameter, wherein the microprocessor is configured to monitor the sensor and provide generate a signal representing the environmental parameter.

15. The fenestration unit of claim 12, wherein the microprocessor is configured for wireless communication with a wireless network accessible to the fenestration unit.

16. The fenestration unit of claim 1, wherein the fenestration unit comprises a window unit.

17. The fenestration unit of claim 1, wherein the fenestration unit comprises a door unit.
